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THESIS PROPOSAL REPORT

Lightweight Edge-Based Deep Learning Framework for Real-Time Plant Disease Detection

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Chapter 1

1. Background and Statement of the Problem

Plant diseases represent one of the primary threats to global food security, agricultural sustainability, and rural economic stability. According to the Food and Agriculture Organization (IPPC Secretariat, 2021), vector-borne crop diseases and pest infestations account for up to 40% of global crop losses annually, costing the global economy over $220 billion. Early and precise identification of plant leaf diseases is essential to allow targeted intervention, minimize chemical pesticide overuse, and prevent catastrophic crop loss.

Traditionally, agricultural disease identification relies on manual visual inspections conducted by experienced farmers, agricultural extension workers, or plant pathologists. However, manual inspection presents severe operational bottlenecks (Barbedo, 2016):

- **Human Error and Subjectivity:** Visual symptoms of different plant pathogens often overlap, leading to misdiagnosis and inappropriate treatment.
- **Scalability Limits:** Manual field surveying across extensive agricultural acreage is labor-intensive, costly, and practically unfeasible for continuous monitoring.
- **Expert Shortages:** In developing agricultural regions, access to certified plant pathologists is severely limited, delaying critical diagnostic responses.

To overcome these constraints, computer vision and Deep Learning (DL)—specifically Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs)—have emerged as powerful solutions for automated visual plant disease diagnosis. Modern deep architectures trained on massive datasets, such as the widely used PlantVillage image collection (Mohanty, Hughes, & Salathé, 2016), achieve image classification accuracy exceeding 95%.

1.1 The Edge Computing Challenge

Despite high theoretical accuracy, deploying standard deep learning models directly to practical farming environments poses a fundamental engineering paradox. Traditional state-of-the-art architectures (such as ResNet-50, DenseNet-121, or VGG-16) feature tens of millions of trainable parameters and require billions of floating-point operations (FLOPs) per image.

Consequently, these models depend on high-performance GPU clusters or cloud-based server infrastructure.

Cloud-dependent agricultural deployment faces critical real-world limitations:

- **Network Connectivity Infrastructure:** Remote agricultural fields in developing nations often lack reliable high-speed cellular networks (3G/4G/5G). Cloud-dependent systems fail completely in offline environments.
- **Latency and Bandwidth Bottlenecks:** Uploading high-resolution leaf images to cloud servers introduces significant latency (often several seconds per image), rendering real-time video stream processing or automated robotic spraying impossible.
- **Data Privacy and Bandwidth Cost:** Continuous video streaming from field cameras to cloud endpoints incurs prohibitive cellular data costs for resource-constrained farmers.

To overcome cloud dependency, the paradigm must shift toward Edge AI—running artificial intelligence inference locally on battery-powered, handheld, or micro-embedded devices (e.g., smartphones, microcontrollers, single-board computers like Raspberry Pi, and hardware accelerators like Google Coral TPU or FPGAs).

However, hardware-constrained edge devices operate under strict physical boundaries: restricted memory (RAM), limited compute power, restricted thermal thresholds, and finite battery capacity. Standard deep learning models rapidly exhaust edge hardware resources, causing system crashes, severe thermal throttling, or unacceptably low frame rates (<2 FPS).

1.2 Statement of the Problem

While lightweight deep learning and model compression techniques (such as post-training quantization, network pruning, and knowledge distillation) have grown rapidly, existing literature demonstrates notable research gaps:

- Most current lightweight models are trained and validated exclusively on clean, controlled, laboratory-style datasets (e.g., PlantVillage; Mohanty et al., 2016), failing significantly when exposed to real-world field conditions (varying sunlight, wind blur, background soil, and leaf occlusion).

- Existing literature heavily emphasizes classification accuracy while neglecting essential edge-hardware execution metrics, including end-to-end latency, throughput (FPS), peak RAM footprint, and live battery power consumption (watts).
- There is a lack of a unified, multi-crop capable, lightweight edge framework capable of delivering real-time performance (>25 FPS) across diverse low-power edge hardware platforms without sacrificing diagnostic accuracy.

Therefore, this research proposes the design, optimization, hardware deployment, and comprehensive field evaluation of an optimized lightweight Edge-AI framework for real-time plant disease detection.

2. Aim and Objectives of the Proposed Research

2.1 Aim

The primary aim of this research is to develop, optimize, deploy, and empirically evaluate a lightweight, multi-crop capable, edge-based Deep Learning framework capable of performing accurate, real-time, low-latency, and energy-efficient plant disease detection directly on hardware-constrained edge devices under real-world agricultural conditions.

2.2 Objectives

To achieve this aim, the proposed research will pursue the following specific objectives:

- **Lightweight Architecture Design:** To design and implement lightweight deep neural network backbones (incorporating depthwise separable convolutions, grouped convolutions, and lightweight attention mechanisms) optimized for low parameter counts (<1.0M for the primary lightweight classification model) and low computational complexity (<0.5 GFLOPs).
- **Model Optimization & Compression:** To evaluate and apply advanced model compression techniques—specifically Post-Training Quantization (PTQ to 8-bit integer formats), weight pruning, and knowledge distillation—to reduce model size substantially while maintaining a target classification accuracy of at least 95%.
- **Multi-Platform Hardware Deployment:** To port, compile, and execute the optimized frameworks across diverse physical edge platforms, including Android smartphones,

ARM Cortex-M microcontrollers, Raspberry Pi 4, and hardware acceleration platforms (Google Coral TPU, NVIDIA Jetson, or FPGA boards).

- **Comprehensive Hardware Benchmarking:** To systematically benchmark the framework's operational performance across both predictive AI metrics (Accuracy, Precision, Recall, F1-score, mAP) and hardware profiling metrics (Inference Latency in ms, Frames Per Second, RAM Memory Footprint in MB, and Power Consumption in watts).
- **Real-World Field Validation & Explainability:** To validate the framework against realistic field-collected datasets featuring adverse lighting and complex backgrounds, while incorporating Explainable AI (Grad-CAM/LIME) to provide visual diagnostic transparency for non-expert farmers.

3. Development / Research Question(s)

3.1 Research Question 1

How can deep neural network architectures be compressed and optimized to minimize memory usage and computational complexity without compromising diagnostic accuracy in complex plant disease identification?

Rationale and Context: Deep neural networks are inherently over-parameterized. Compression methods like 8-bit quantization reduce precision from 32-bit floating-point (FP32) to 8-bit integers (INT8), dramatically cutting memory bandwidth requirements. However, aggressive compression can cause drop-offs in classification accuracy. This question investigates the exact mathematical trade-off between model compression ratios, parameter scaling, and classification performance.

3.2 Research Question 2

Which combination of lightweight deep learning architecture, compiler runtime, and edge hardware accelerator achieves the optimal balance between inference speed (FPS), thermal efficiency, and battery power consumption?

Rationale and Context: Edge hardware architectures vary widely—ranging from general-purpose CPUs and microcontrollers to dedicated Tensor Processing Units (TPUs) and FPGAs.

Similarly, software compilation backends (TensorFlow Lite, ONNX Runtime, OpenVINO) handle hardware execution differently. Determining the optimal hardware-software pairing is crucial for building practical, low-cost field devices.

3.3 Research Question 3

How robust is the optimized edge-based framework when deployed under unpredictable, real-world field conditions, and how can visual Explainable AI (XAI) improve user trust among agricultural end-users?

Rationale and Context: Models trained on clean laboratory images often fail outdoors due to shadows, glare, post-rain reflections, and background noise. Evaluating field robustness ensures practical reliability. Furthermore, visual heatmaps (Grad-CAM) allow farmers to verify that the model is focusing on actual disease lesions rather than background artifacts.

Chapter 2

4. Literature Review

The development of automated plant disease detection has progressed through three main technological generations: classical machine learning with handcrafted features, deep learning using heavy convolutional networks, and edge-native AI deployment using optimized lightweight architectures. This section critically analyzes 15 relevant research studies.

4.1 Comprehensive Analysis of Literature

4.1.1 Quantization and Model Compression Strategies

Model quantization is a central strategy for edge deployment. Khan et al. (2023) developed a plant disease detection model using MobileNetV3-Small and demonstrated that Post-Training Quantization (PTQ) can reduce the parameter footprint from about 1.5 million to 0.93 million while maintaining 99.50% classification accuracy. This result shows the potential of INT8-oriented compression for resource-constrained devices. However, the study primarily relied on PlantVillage images, so robustness under complex outdoor backgrounds remains an open issue.

De Vita et al. (2021) investigated Dynamic K-Means Weight Compression for plant disease detection and reported about 85.2% memory reduction while maintaining approximately 95.0% accuracy. Although this is valuable for memory-constrained deployment, the study does not fully address the combined evaluation of real-time throughput, energy consumption, and field robustness on multiple physical edge platforms.

4.1.2 Comparative Hardware Benchmarking & Acceleration

Wei, Al Riza, and Nugroho (2022) compared AlexNet, VGG16, ResNet50, and DenseNet121 using the PlantVillage dataset and evaluated the models across CPU, GPU, and VPU environments. DenseNet121 achieved 96.4% accuracy and showed that edge hardware and software runtimes can materially affect practical deployment performance. Nevertheless, the relatively large model size motivates further investigation of more compact architectures.

Majeed, Ojo, and Zahid (2024) evaluated GoogleNet, MobileNetV2, EfficientNet-B0, ResNet-18, SqueezeNet, and NasNetMobile on Raspberry Pi 4, Raspberry Pi 4 with Google Coral TPU, NVIDIA Jetson Orin Nano, and NVIDIA Jetson AGX Orin. GoogleNet and MobileNetV2

reached up to 98.25% accuracy, while the Raspberry Pi + Google Coral configuration provided the best reported cost/FPS ratio of 0.14. This study strongly supports the need to evaluate both model quality and hardware efficiency.

Hoang et al. (2026) presented a comprehensive comparison of ShuffleNetV2, MobileNetV3-Small, SqueezeNet, MobilePlantViT, DenseNet121, ResNet50, and VGG16 for tomato disease classification. The study reported that MobilePlantViT offered a strong balance between accuracy and efficiency and also evaluated Grad-CAM, SHAP, and LIME using a perturbation stability score. However, the evaluation was still largely based on controlled datasets and CPU measurements, leaving broader physical edge-device and field validation opportunities.

4.1.3 Real-Time Detection & Object Localization

For field robotics and handheld scanners, disease detection must occur in continuous video streams. Sun et al. (2021) developed MEAN-SSD, combining an Apple-Inception module with Single Shot MultiBox Detector (SSD) for 5 apple leaf diseases. MEAN-SSD achieved a mean Average Precision (mAP) of 83.12% at 12.53 FPS on mobile hardware.

Zhao et al. (2026) proposed HGS-YOLO, a lightweight YOLOv11-based tomato leaf disease detector using HGNetV2, HS-FPN with channel attention, and MPDIoU loss. On the reported benchmark, the model achieved 93.6% mAP@50, 1.3M parameters, 25.0 ms latency, 40.0 FPS, and approximately 9.8 W during indoor deployment testing on a D-Robotics RDK X5. A small outdoor greenhouse test also demonstrated operational feasibility, although broader multi-site validation remains necessary.

4.1.4 Ultra-Low-Power Microcontrollers and FPGA Hardware

For extremely low-cost deployment, microcontrollers and customized hardware accelerators are required. Chew Jing Xan et al. (2024) focused on deploying lightweight CNNs (FD-MobileNet) onto ARM Cortex-M microcontrollers for rice leaf disease detection. FD-MobileNet achieved 98.44% accuracy with low computational overhead.

At the silicon-hardware level, Luo et al. (2023) developed LiteCNN with 0.025M parameters and 0.0056 GFLOPs, using knowledge distillation and lightweight convolutional design. On a ZYNQ Z7-Lite 7020 FPGA, LiteCNN achieved 95.71% accuracy, about 0.071 seconds per frame, and 2.41 W power consumption. This demonstrates the value of co-optimizing model architecture and hardware acceleration.

4.1.5 Novel Architectures and Attention Mechanisms

Ashwinkumar et al. (2022) proposed an optimal MobileNet-based CNN approach (OMNCNN) using image preprocessing, MobileNet feature extraction, and optimization, reporting 98.7% accuracy. Chen et al. (2022) proposed MS-DNet, combining depthwise separable convolutions, DenseNet transition structures, SE attention, and progressive training; the model achieved an average accuracy of 98.32%. These studies demonstrate that compact architectures and attention mechanisms can preserve high predictive performance while reducing computational requirements.

Remya et al. (2026) introduced an Adaptive Multi-Scale Bilinear Fusion Network (AMSBFN) that combines MobileNetV3 and EfficientNet-B0 with spatial-channel attention and bilinear fusion. Evaluated on 33,826 PlantVillage tomato images across 10 classes, AMSBFN achieved 97.47% classification accuracy and used Grad-CAM++ for visual interpretation. Its controlled-dataset setting nevertheless leaves room for independent field validation.

4.1.6 Smartphone Field Deployment and Explainability

Cui et al. (2026) introduced LMRNet for mountain rice disease recognition on edge devices. The model contains approximately 0.90M parameters and 0.54 GFLOPs; field validation on Android smartphones achieved 92.41% average accuracy and approximately 22.47 FPS. The reported degradation under backlight, occlusion, and post-rain glare highlights the importance of field-oriented evaluation.

Al Kafi et al. (2026) developed LeafAI, a two-stage Logistic Regression + MobileNetV3 framework with Grad-CAM explainability. On a large-scale test of 1,227 images, the hybrid approach reduced total inference time from 4,548 seconds to 1,010.13 seconds, corresponding to a substantial speed improvement with a small reported accuracy trade-off. The work strengthens the case for combining efficiency with interpretability, while further validation across diverse crops and physical edge hardware is still needed.

4.2 Comparative Literature Matrix

The table below provides a structured comparative summary of all 15 reviewed papers (Table 1).

Table 1. Comparative Literature Matrix

Ref.	Authors / Year	Model / Architecture	Optimization / Technique	Deployment / Evaluation	Key Result	Research Gap
1	Khan et al. (2023)	MobileNetV3-Small	PTQ (FP32→INT8)	Edge/mobile setting	99.50% accuracy; parameters 1.5M→0.93M	Mainly PlantVillage; field robustness not established.
2	Wei et al. (2022)	DenseNet121	Downsampling; OpenVINO	CPU/GPU/VPU	96.4% accuracy	Dense model is less suitable for very low-resource devices.
3	Sun et al. (2021)	MEAN-SSD	Lightweight CNN + SSD	Mobile deployment	83.12% mAP; 12.53 FPS	Focused on five apple diseases.
4	Chew Jing Xan et al. (2024)	FD-MobileNet	Quantization / lightweight optimization	ARM-M microcontroller	98.44% accuracy	Limited crop and field-environment coverage.
5	Majeed et al. (2024)	GoogleNet / MobileNetV2 and other CNNs	Edge-device benchmarking	Raspberry Pi, Coral TPU, Jetson	Up to 98.25% accuracy; Coral cost/FPS 0.14	Comparative study rather than a unified multi-crop framework.
6	Hoang et al. (2026)	MobilePlantViT + lightweight CNNs	Model/CPU benchmarking + XAI	CPU edge feasibility	MobilePlantViT ~99.4% accuracy; XAI stability evaluated	Physical edge-device and field validation remain limited.
7	Ashwinkumar et al. (2022)	OMNCNN	MobileNet + optimization	CNN-based classification	98.7% accuracy	Robustness under complex backgrounds needs further study.
8	Chen et al. (2022)	MS-DNet	Depthwise separable conv.; SE attention	Mobile-oriented classification	98.32% average accuracy	Generalization across broader field conditions remains open.
9	Luo et al. (2023)	LiteCNN	Knowledge distillation; lightweight convolution	ZYNQ FPGA	95.71% accuracy; 0.071 s/frame; 2.41 W	Hardware-specific optimization limits direct portability.
10	De Vita et al. (2021)	Deep neural network + K-Means compression	Dynamic K-Means weight compression	Edge deployment study	~95% accuracy; ~85.2% memory reduction	Runtime FPS and power were not comprehensively benchmarked.
11	Cui et al. (2026)	LMRNet	Lightweight CNN design	Android smartphones	92.41% field accuracy; ~22.47 FPS	Performance drops under backlight, occlusion and glare.
12	Tonmoy et al. (2025)	MobilePlantViT	Hybrid CNN–ViT design	Mobile/edge-oriented research	0.69M parameters; strong cross-dataset performance	Primarily research-stage evaluation; broader hardware validation needed.
13	Zhao et al. (2026)	HGS-YOLO	HGNetV2 + HS-FPN + CA + PTQ	D-Robotics RDK X5	93.6% mAP@50; 40 FPS; 25 ms; 9.8 W	Broader multi-site, seasonal field validation is still needed.
14	Remya et al. (2026)	AMSBFN	Bilinear fusion + spatial-channel attention	Edge-oriented tomato classification	97.47% accuracy	Evaluation mainly on PlantVillage tomato images.
15	Al Kafi et al. (2026)	LeafAI (LR + MobileNetV3)	Two-stage inference + Grad-CAM	Entry-level laptop / edge-oriented	1,227-image test: 4,548 s → 1,010.13 s	Needs broader crops, datasets and physical edge-hardware validation.

4.3 Research Gap and Synthesis

A synthesized critical review of the existing literature highlights three major research gaps:

- **The Laboratory-to-Field Performance Gap:** Most existing studies report accuracies above 98% on benchmark datasets like PlantVillage. However, these datasets feature uniform backgrounds and static lighting. When deployed in outdoor agricultural fields, factors such as solar glare, moving shadows, leaf overlap, and soil backgrounds degrade diagnostic accuracy.
- **Incomplete Hardware Benchmarking:** A significant portion of existing research evaluates predictive accuracy while omitting hardware profiling. In practical agricultural deployments, latency, throughput (FPS), RAM allocation, and thermal/power consumption are equally important success factors.
- **Lack of Crop Generalization:** Current state-of-the-art lightweight frameworks are heavily tailored toward single crops (e.g., tomato or rice only). There is a distinct need for an optimized, multi-crop edge framework that supports multiple plant species simultaneously while operating within strict low-power constraints. The proposed study will evaluate a selected set of representative crops rather than claiming universal generalization to all plant species.

Chapter 3

5. Research Methodology

To address these research gaps, the proposed methodology follows a systematic, four-phase approach.

5.1 Phase 1: Dataset Curation and Field Data Augmentation

- **Dataset Integration:** A composite dataset will be created by combining benchmark laboratory datasets (PlantVillage) with outdoor field images representing major economic crops (tomato, rice, apple, and local crops).
- **Field Condition Augmentation:** Data pipelines will introduce synthetic environmental perturbations—including varying color temperatures (direct sunlight vs. cloud cover), simulated wind motion blur, shadows, and background noise (soil, weeds, farming equipment)—to ensure real-world model robustness.

5.2 Phase 2: Lightweight Model Architecture and Compression

- **Base Architecture Selection:** Lightweight convolutional backbones (MobileNetV3-Small, ShuffleNetV2) and object detection backbones (HGS-YOLO) will serve as baseline architectures.
- **Structural Modifications:** Models will incorporate depthwise separable convolutions, linear bottlenecks, and spatial-channel attention modules (Squeeze-and-Excitation / Convolutional Block Attention Module).

Model Compression Pipeline:

- **Post-Training Quantization (PTQ):** Converting weights and activation tensors from 32-bit floating-point (FP32) to 8-bit integer (INT8) representation.
- **Structured Weight Pruning:** Removing redundant convolutional filters based on L1-norm thresholds to minimize FLOPs.
- **Knowledge Distillation:** Training lightweight “student” models using soft labels generated by heavy “teacher” networks (e.g., ResNet-50).

5.3 Phase 3: Hardware Compilation and Porting

- **Runtime Conversion:** The optimized neural network models will be exported to optimized runtime formats, including TensorFlow Lite (.tflite), ONNX Runtime (.onnx), and Intel OpenVINO Intermediate Representation (.xml/.bin).

Deployment Platforms: Models will be ported to four representative edge hardware tiers:

- Tier 1: Commercial Android Smartphones (ARM Cortex CPUs + Adreno/Mali GPUs).
- Tier 2: Single-Board Computers (Raspberry Pi 4 Model B).
- Tier 3: Edge AI Accelerators (Raspberry Pi 4 + Google Coral USB Accelerator TPU).
- Tier 4: Low-Power Microcontrollers (ARM Cortex-M series).

5.4 Phase 4: Empirical Benchmarking, Field Validation and Explainable AI

- **Benchmarking Protocol:** The optimized models will be evaluated using accuracy, precision, recall, F1-score, mAP@50 (for detection models), parameter count, model size, GFLOPs, RAM usage, latency, FPS, and power consumption.
- **Field Validation:** A separate field-oriented test set will be collected or curated under different lighting, backgrounds, leaf orientations, partial occlusions, and motion conditions to measure the laboratory-to-field performance gap.
- **Explainable AI:** Grad-CAM or a comparable visual XAI method will be applied to selected classification models to verify whether predictions focus on disease-relevant leaf regions. Explanation quality and failure cases will be documented alongside model performance.
- **Comparative Analysis:** Results will be compared across the selected edge platforms to identify the best accuracy–latency–memory–power trade-off and to determine the most practical deployment configuration.

6. Timetable or Work-plan for the Proposed Study

The research project is structured across a six-month implementation period. The work plan is organized sequentially so that each stage produces inputs required for the next stage, while allowing limited overlap for model refinement, testing, and documentation.

Table 2. Timetable / Work Plan (Gantt Chart)

Research Task / Milestone	Sep 2026	Oct 2026	Nov 2026	Dec 2026	Jan 2027	Feb 2027
Task 1: Literature Review & Gap Identification	●					
Task 2: Dataset Curation & Augmentation Pipelines	●	●				
Task 3: Lightweight Model Design, Training & Compression		●	●	●		
Task 4: Edge Porting & Runtime/Compiler Optimization				●	●	
Task 5: Benchmarking, XAI & Field Validation					●	●
Task 6: Thesis Documentation, Revision & Defense Preparation					●	●

7. Outcomes of the Proposed Research

The expected deliverables and research contributions of this project are:

- **An Optimized Lightweight Framework:** A lightweight deep learning model featuring <1.0M parameters and <0.5 GFLOPs, optimized for edge-device deployment.
- **Cross-Platform Deployment Runtime:** Fully compiled runtime binaries (.tflite, .onnx) ready for deployment on Android smartphones, Raspberry Pi devices, microcontrollers, and hardware accelerators.
- **High-Speed Real-Time Performance:** A functional real-time diagnostic prototype targeting ≥25 FPS and ≤30 ms end-to-end inference latency on the selected handheld edge platform. Performance will be reported separately for each tested hardware configuration rather than generalized across all devices.
- **Energy-Efficient Offline Diagnostic Utility:** A low-power deployment solution capable of operating offline on battery-powered agricultural devices, reducing dependence on cloud connectivity and enabling timely on-site diagnosis.

- **Empirical Edge-AI Benchmark Suite:** A comprehensive dataset and benchmarking report detailing the performance trade-offs between predictive accuracy, latency, memory footprints, and power consumption across edge platforms.
- **Research and Practical Contribution:** The project will provide a reproducible evaluation protocol linking model accuracy with parameter count, computation, latency, memory, energy use, deployment format, and explainability. The resulting prototype and benchmark evidence are intended to support future low-cost agricultural AI systems in Bangladesh and similar resource-constrained environments.

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